

A Lightweight Multi-Modal Tiny LLM Framework for Privacy-Constrained Academic Assistance in University Environments

Draft manuscript - evidence-aligned version

Abstract

Universities increasingly need local academic assistants that can answer questions from course material while preventing protected assessment artifacts from leaking to students. We present a lightweight, CPU-oriented framework that separates public learning corpora from protected exam corpora, supports auditable PDF/text ingestion with an optional OCR-backed image path, performs Bloom-taxonomy classification for moderation, and applies role-aware query and output screening before student-facing generation. The paper does not claim formal differential privacy. Instead, it evaluates bounded empirical resistance under an explicit university threat model. On the Figshare Bloom exam dataset, the selected classifier reaches 0.769 test accuracy and 0.744 macro-F1 after duplicate/conflict auditing. Cross-dataset transfer with MoocRadar shows large exact-label degradation, indicating that Bloom classifiers remain domain-sensitive; ordinal metrics expose where severe cognitive-level jumps are reduced or amplified. An expanded role-aware privacy evaluation blocks all evaluated student reconstruction attacks, allows 0.857 of benign student study prompts, and allows teacher moderation prompts. A small offline QA/RAG evaluation reports answer F1, retrieval hit rate, and unsupported-answer risk; Qwen2.5-1.5B-Instruct-Q4_K_M.gguf is then evaluated over the same retrieved contexts. The image pipeline evaluation reports OCR readiness or measured OCR quality depending on the installed backend. These results support a conservative claim: local role separation plus leakage screening is a practical deployment pattern for privacy-constrained academic assistance, but not a proof of universal privacy or cross-domain Bloom generalization.

1. Introduction

RAG-based educational assistants can help students ask questions over lecture PDFs, images, and notes, while helping teachers moderate exam questions and label Bloom levels. In a university environment these two workflows must not share the same retrieval surface: a student should not be able to recover protected exam uploads by asking for previous documents, summaries, paraphrases, or topic lists. This creates a deployment gap between general privacy-preserving RAG research and education-specific Bloom-classification systems.

The central research question is: how far can a lightweight local framework go in combining multimodal academic ingestion, Bloom-aware moderation, and role-aware privacy controls without overclaiming formal security? We answer with a system and evaluation protocol that makes failures visible: exact Bloom transfer degrades under domain shift, privacy is evaluated with attack prompts rather than asserted, and protected assessment material is handled as a teacher-only resource.

Contributions

- A deployment pattern for privacy-constrained educational RAG that separates public student retrieval from protected teacher moderation.
- An evaluation protocol spanning answer correctness, retrieval quality, unsupported-answer risk, role-aware privacy attacks, Bloom robustness, and image/OCR ingestion.
- A Bloom-classification analysis framed as evaluation insight: exact cross-domain labels are brittle, while ordinal error metrics reveal usable moderation risk structure.
- A conservative claim policy that distinguishes taxonomy-based empirical leakage resistance from formal privacy guarantees.

2. Related Work and Gap

Prior RAG work establishes the utility of retrieval augmentation and also documents privacy risks in retrieval and generation. Recent privacy-preserving RAG work explores differential privacy, query protection, cloud RAG, tabular settings, and multimodal leakage. In parallel, educational NLP work has improved automated Bloom classification for exam questions. The gap addressed here is not a new cryptographic mechanism; it is the integrated university setting where public study material, protected exams, student access, teacher moderation, and Bloom labeling coexist in one lightweight local workflow.

3. System and Threat Model

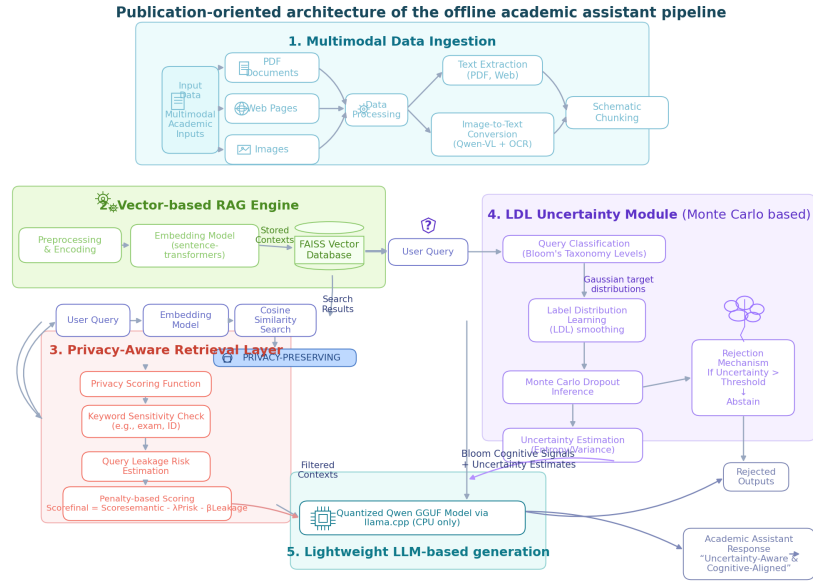


Figure 1. Role-separated local architecture. Public material is indexed for student RAG; protected exam material is routed through teacher-only moderation and student-facing leakage screening.

The attacker is a student with black-box access to the assistant. The attacker may ask for uploaded documents, paraphrases, summaries, exact questions, partial spans, or semantically equivalent practice items. The teacher/moderator role is trusted to view protected exam material for moderation and Bloom labeling. The framework targets bounded reconstruction resistance and operational separation; it does not defend against compromised teacher accounts, database exfiltration, side channels, or formal membership inference.

4. Datasets and Protocol

Dataset	Use	Rows after audit	Notes
Figshare Bloom exam questions	Primary in-domain Bloom classification	2330	Exact duplicates removed; conflicting duplicate labels removed.
MoocRadar problem set	External cross-domain Bloom transfer	9324	Used to test domain shift, not to claim universal generalization.
Synthetic protected exam micro-corpus	Privacy-guard attack and utility probes	125	Small, controlled corpus for policy evaluation; not a real deployment study.
Offline academic QA set	Answer correctness and retrieval grounding; extractive and Qwen-generated	10	Small local QA benchmark; external HotpotQA/NQ-style evaluation remains future work.
Synthetic typed image set	OCR/image ingestion path	3	Reports OCR quality when a backend is installed; otherwise reports backend unavailability.
Generated PDF/image RAG smoke set	End-to-end file ingestion and extractive RAG	2	Tests PDF RAG and OCR-backed image RAG as separate pipeline cases.

Dataset usage is therefore aligned with claim scope: Figshare supports in-domain Bloom evidence, MoocRadar supports cross-domain stress testing, the protected micro-corpus supports guard-behavior analysis, the offline QA set tests grounding behavior, and the synthetic image set tests whether the OCR ingestion path is measurable. The present artifacts do not support claims about large-scale real university deployment or formal privacy.

5. Results

5.1 In-domain Bloom classification

Metric	Value
Accuracy	0.769
Macro-F1	0.744
Weighted-F1	0.768

Metric	Value
Mean ordinal error	0.426
Within-one-level accuracy	0.903
Severe ordinal error	0.097

5.2 Domain shift

Protocol	Setting	Model	Macro-F1	Acc.	Within-1	Severe
binary Bloom transfer	Figshare in-domain	linear_svm_balance d	0.907	0.916	1.000	0.000
binary Bloom transfer	MoocRadar in-domain	logreg_balanced	0.759	0.766	1.000	0.000
binary Bloom transfer	Figshare -> MoocRadar	linear_svm_balance d	0.379	0.560	1.000	0.000
binary Bloom transfer	MoocRadar -> Figshare	linear_svm_balance d	0.471	0.471	1.000	0.000
ternary Bloom transfer	Figshare in-domain	logreg_balanced	0.858	0.873	0.959	0.041
ternary Bloom transfer	MoocRadar in-domain	logreg_balanced	0.698	0.743	0.940	0.060
ternary Bloom transfer	Figshare -> MoocRadar	linear_svm_balance d	0.248	0.458	0.882	0.118
ternary Bloom transfer	MoocRadar -> Figshare	linear_svm_balance d	0.261	0.312	0.935	0.065

Cross-Domain Bloom Performance

scheme	setting	model	accuracy	macro_f1	mean_ordinal_error	within_one_level_accuracy	severe_error_rate	gate_050_human_loop	gate_050_accepted_severe_error	gate_050_severe_error_caught
binary	within_dataset_figshare	linear_svm_balanced	0.916	0.907	0.084	1.000	0.000	0.398	0.000	0.000
binary	within_dataset_moocradar	logreg_balanced	0.766	0.759	0.234	1.000	0.000	0.728	0.000	0.000
binary	figshare_to_moocradar	linear_svm_balanced	0.560	0.379	0.440	1.000	0.000	0.926	0.000	0.000
binary	moocradar_to_figshare	linear_svm_balanced	0.471	0.471	0.529	1.000	0.000	0.931	0.000	0.000
ternary	within_dataset_figshare	logreg_balanced	0.873	0.858	0.167	0.959	0.041	0.440	0.004	0.947
ternary	within_dataset_moocradar	logreg_balanced	0.743	0.698	0.317	0.940	0.060	0.406	0.022	0.780
ternary	figshare_to_moocradar	linear_svm_balanced	0.458	0.248	0.660	0.882	0.118	0.986	0.062	0.993
ternary	moocradar_to_figshare	linear_svm_balanced	0.312	0.261	0.753	0.935	0.065	0.946	0.000	1.000

Figure 2. Cross-domain Bloom performance. Exact transfer degrades sharply, so the claim is cognitive robustness analysis rather than solved generalization.

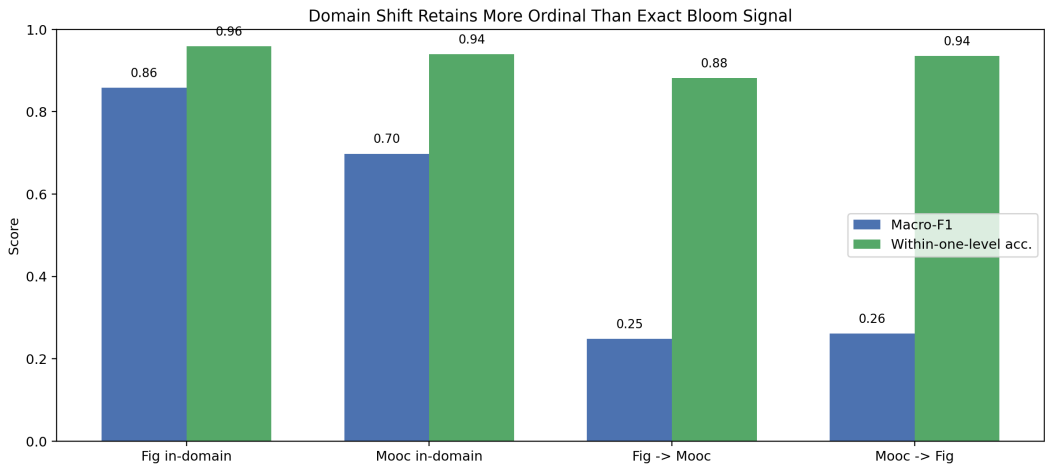


Figure 3. Ordinal analysis under domain shift. Within-one and severe-error metrics expose structure hidden by exact accuracy.

5.3 Role-aware privacy guard

Metric	Value	Interpretation
Student attack block rate	1.000	Measured on the defined prompt taxonomy only.

Metric	Value	Interpretation
Student benign allow rate	0.867	Ordinary study help is mostly preserved; blocked cases motivate interface wording.
Teacher moderation allow rate	1.000	Trusted moderation workflow remains available.
Attack prompts	104	Expanded synthetic adversarial set, still below a full external red-team study.
Total privacy rows	125	Includes attacks, benign student prompts, and teacher moderation prompts.

The privacy guard evidence is promising but bounded. A high venue version should emphasize that the expanded prompt evaluation is an attack taxonomy, not a proof. The student benign allow rate below 1.0 is also useful: it reveals the utility cost of strict protected-artifact screening and motivates future adaptive policies.

5.4 QA/RAG grounding

System	Token-F1	EM	Hit@1	Hit@3	Unsupported
Qwen2.5-1.5B-Q4_K_M	0.914	0.800	0.900	1.000	0.000
Proposed	0.973	0.900	0.900	1.000	0.000
VanillaRAG	0.973	0.900	0.800	1.000	0.000
BM25	0.973	0.900	0.900	1.000	0.000
NoRAG	0.025	0.000	0.000	0.000	1.000

This QA/RAG evaluation is deliberately small and offline. It measures answer correctness, support-document retrieval, and an unsupported-answer proxy. The primary generator row uses Qwen2.5-1.5B-Instruct-Q4_K_M.gguf through llama.cpp with greedy decoding; the extractive rows remain as diagnostic controls. It should be replaced or supplemented with HotpotQA, Natural Questions, DocVQA, or instructor-authored course QA before making broad utility claims.

5.5 OCR and image pipeline

Metric	Value
Qwen model	Qwen2.5-1.5B-Instruct-Q4_K_M.gguf
Qwen model file MB	986.049
Qwen PDF RAG	True
Qwen image RAG	True
PDF RAG smoke test	True
Image RAG smoke test	True
RAG answer accuracy on OK cases	1.000
Backend	pytesseract
Backend available	True
Samples evaluated	3
Mean OCR token-F1	0.963
Mean word error rate	0.048
Access-level preservation	1.000
Modality preservation	1.000

The multimodal claim is therefore scoped to an implemented and measurable image ingestion path plus a PDF RAG path, both evaluated with Qwen over retrieved context. PDF RAG is tested with native text extraction. Image RAG is OCR-backed: if no OCR backend is installed, the result is reported as missing backend evidence rather than image-understanding performance. Scanned handwriting, complex layouts, and DocVQA-style reasoning remain outside the current evidence.

6. Discussion

The results do not show a universally robust Bloom classifier or a formally private RAG system. They do show a publishable systems contribution if framed around the university deployment gap: local resource-aware operation, explicit role separation, protected

assessment handling, and evaluation that reports failures. The strongest finding is that exact cognitive-label transfer is brittle, while ordinal metrics and guarded deployment policies provide a safer basis for moderation support.

7. Limitations and Required Next Evidence

- Privacy is not differential privacy; add DP/query-privacy baselines only if the implementation actually includes them.
- The privacy attack set is synthetic and should be expanded with external red-team prompts before submission.
- The multimodal image/PDF path needs OCR-quality reporting if the paper foregrounds multimodality.
- The QA/RAG benchmark is small and local; replace or supplement it with HotpotQA, Natural Questions, DocVQA, or instructor-authored course QA before broad claims.
- A teacher/user study or expert moderation agreement study would strengthen claims for higher venues.

8. Conclusion

A lightweight university assistant can be made more publishable by narrowing the claim: not a general privacy solution, but a reproducible local framework that separates public learning RAG from protected exam moderation and evaluates the result under domain-shift and leakage probes. This framing fills a real gap between general privacy-preserving RAG and educational Bloom-classification work.

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